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SOFTWARE DESIGN DOCUMENT (SDD) for

<Online Doctor Appointment Application>

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S23BINFT1M01088

BSIT 7th (2M)

Session: FALL 2023 – 2027

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Revision History

Date	Description	Author	Comments
<date>	<Version 1>	<Your Name>	<First Revision>

Document Approval

The following Software Design Document (SDD) has been accepted and approved by the following:

Signature	Printed Name	Title	Date
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Table of Contents

1. Introduction	3
1.1. Purpose.....	3
1.2. Scope	4
1.3. Overview	4
1.4. Reference Material	5
1.5. Definitions and Acronyms	5
2. System Overview	6
3. System Architecture	7
3.1. Architectural Design	7
3.2. Decomposition Description	8
3.3. Design Rationale	8
4. Data Design	9
4.1. Data Description.....	9
4.2. Data Dictionary.....	9
5. Component Design	10
6. Human Interface Design	13
6.1. Overview of User Interface	13
6.2. Screen Images	15
6.3. Screen Objects and Actions	17
7. Requirements Matrix	17
8. Appendices	18

1. INTRODUCTION

1.1. PURPOSE

THE INTENDED AUDIENCE FOR THIS DOCUMENT INCLUDES ALL STAKEHOLDERS INVOLVED IN THE ONLINE DOCTOR APPOINTMENT APPLICATION (ODAA) PROJECT, SUCH AS ADMINISTRATIVE STAFF, DOCTORS, PATIENTS, AND DEVELOPERS. IT IS RECOMMENDED THAT THEY REFER TO THIS DOCUMENT FOR THE OFFICIAL AND UP-TO-DATE REQUIREMENTS. ANY REQUIREMENTS MENTIONED ELSEWHERE, EITHER IN WRITTEN OR VERBAL FORM, SHOULD NOT BE CONSIDERED VALID UNLESS THEY APPEAR IN THIS DOCUMENT OR ITS REVISIONS.

1.1 Overview

This section provides a general description of the Online Doctor Appointment Application (ODAA) and its purpose.

1.2 User Characteristics

Describes the different types of users who will interact with the ODAA and their roles, such as administrative staff, doctors, and patients.

1.3 General Constraints

Specifies any constraints or limitations that need to be considered during the development of the ODAA.

1.4 Assumptions

States any assumptions made during the requirements gathering process and their impact on the ODAA.

User and Application Requirements

This section outlines the specific requirements of the ODAA, including both functional and nonfunctional requirements.

Design Documentation and Implementation Decisions

Provides an overview of the design documentation and implementation decisions made for the ODAA.

Test Cases

Presents a set of test cases to verify and validate the functionality of the ODAA.

Progress and Challenges

Summarizes the progress made during the development of the ODAA and discusses any challenges faced during the process.

Conclusion

Concludes the Software Requirements Specification (SRS) document and emphasizes its importance as the primary source of requirements for the project.

1.2. SCOPE

The proposed software product is the Online Doctor Appointment Application (ODAA). In this project we are going to design and build a fully functional application based online doctor appointment system. The aim of this project is to create an online doctor appointment Application. Customers can be able to book doctor's appointment anywhere in the world, this can be done via Application. Doctors can also register to this Application, view appointments and set available time for appointment thereby making it more convenient for them. Administrator also have access to the Application and able to change information of the Application and have access to database. Payment is made after appointment.

1.3. OVERVIEW

This Software Requirements Specification (SRS) is the requirements work product that formally specifies Online Doctor Appointment Application (ODAA). It includes the results of both business analysis and systems analysis efforts. Various techniques were used to elicit the requirements and we have identified your needs, analyzed and refined them. The objective of this document therefore is to formally describe the system's high level requirements including functional requirements, non-functional requirements and constraints.

The detail structure of this document is organized as follows:

First we will provide an overview of the business domain that the proposed Online Doctor Appointment Application (ODAA) will support. These include a general description of the product, user characteristics, general constraints, and any assumptions for this Application. Second we will mention the User and Application Requirements. After that, we will take a look at the design documentation and implementation decisions of the Application and some test cases. At the end, my report will conclude the progress I made and discuss the problems faced during the work.

1.4. REFERENCE MATERIAL

Google search engine

1.5. DEFINITIONS AND ACRONYMS

Table : 1 Term And Definition

Term	Definition
Appointment	An arrangements to meet someone at a particular time and place.
Android Application	An application run on the internet.
Fire-base	Collection of all the information monitored by application.
Constraints	Lamination of application.
Business Domain	A set of classes that represents object in the business model being implemented.

Self-contained	Having all that is needed by it self.
Administrative staff	Person responsible for management of operations.
Graphic user interface	A visual way of interacting with a computer using items.
Software Requirements Specification	A document that completely describes all of the functions of a proposed application and constraints under which must operate.
Stakeholder	Any person with an interest in the project who is not developer
User	Patient, Doctor and Admin

2. SYSTEM OVERVIEW

This Software Requirements Specification (SRS) is the requirements work product that formally specifies Online Doctor Appointment Application (ODAA). It includes the results of both business analysis and systems analysis efforts. Various techniques were used to elicit the requirements and we have identified your needs, analyzed and refined them. The objective of this document therefore is to formally describe the system's high level requirements including functional requirements, non-functional requirements and constraints.

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2.2 PRODUCT FUNCTIONS

- a. Provide an application that enables patients to book an appointment with any available doctor.
-

- b. Doctors will be able to view their appointments and manage them properly.
- c. Coordinate various calendars and finding available time slots for appointment.
- d. Reserve equipment and rooms for appointment.
- e. Alert patients in case there is an earlier available time slot.

2.3 USER CHARACTERISTICS

The system will be used in the hospital. The doctors and patients will be the main users. Given the condition that not all the users are computer-literate. Some users may have to be trained on using the application. The system is also designed to be user-friendly. It uses a Graphical User Interface (GUI).

Patient: These are the people who want to make the appointment.

Doctor: These are the specialist whom appointments are being booked for.

Database Administrator: They are responsible for maintaining and overseeing the fire base of the application.

2.4 GENERAL CONSTRAINTS

The system must be delivered by the proposed deadline.

The system must be user-friendly. **2.5**

ASSUMPTIONS AND DEPENDENCIES

It is assumed that Android device will be available before the application is installed and tested.

It is assumed that the Hospital will have enough trained staff to take care of the application.

3. System Architecture

3.1. ARCHITECTURAL DESIGN

Develop a modular program structure and explain the relationships between the modules to achieve the complete functionality of the system. This is a high level overview of how responsibilities of the system were partitioned and then assigned to subsystems. Identify each high level subsystem and the roles or responsibilities assigned to it. Describe how the subsystems collaborate with each other in order to achieve the desired functionality. Don't go into too much detail about the individual subsystems. The main purpose is to gain a general understanding of how and why the system was decomposed, and how the individual parts work together. Provide a diagram showing the major subsystems and data repositories and their interconnections. Describe the diagram if required.

3.2. DECOMPOSITION DESCRIPTION

Provide a decomposition of the subsystems in the architectural design. Supplement with text as needed. You may choose to give a functional description or an object oriented description.

For a functional description, put top level data flow diagram (DFD) and structural composition diagrams. For an Object-Oriented (OO) description, put subsystem model, object diagrams, generalization hierarchy diagram(s) (if any), aggregation hierarchy diagram(s) (if any), interface specifications, and sequence diagrams here.

3.3. DESIGN RATIONALE

Discuss the rationale for selecting the architecture described in section 3.1 including critical issues and trade/offs that were considered. You may discuss other architectures that were considered, provided that you explain why you didn't choose them.

3.4 DATA FLOW DIAGRAMS (DFD)

Data Flow diagram for Online Doctor Appointment Application

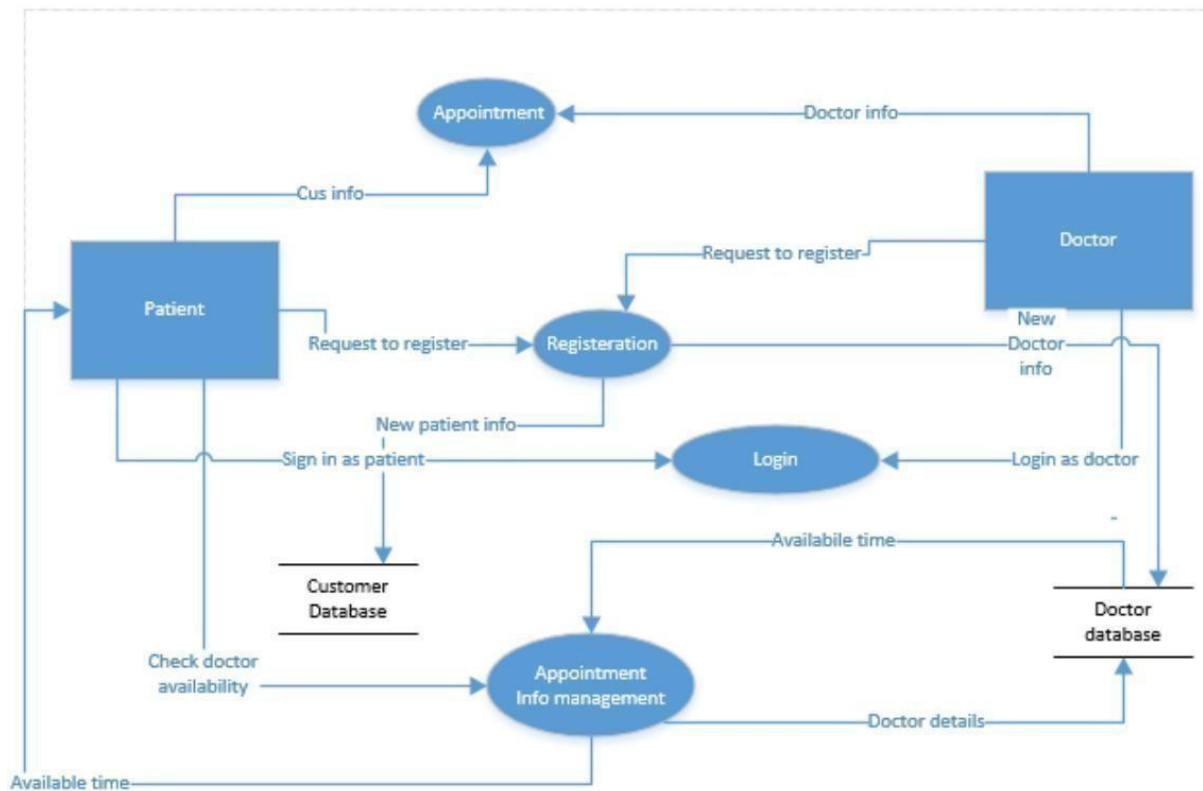


Figure 1. Data Flow diagram for Online Doctor Appointment Application

4. DATA DESIGN

4.1. DATA DESCRIPTION

Explain how the information domain of your system is transformed into data structures.

Describe how the major data or system entities are stored, processed and organized. List any databases or data storage items.

4.2. DATA DICTIONARY

Table : 2 Admin &User:

Name	Description	Data Type	Length	Null
ID	Admin ID	Integer	5	NO
Name	Admin Name	String	80	NO
Address	Admin Address	String	100	NO
MobileNo	Mobile Number	String	12	NO
Username	Username for Login	String	40	NO
Password	Password for Login	Integer	8	NO
ID	Admin ID	Integer	5	NO

Name	Description	Data Type	Length	Null
User ID	User ID	Integer	5	NO
Name	User Name	String	80	NO
Address	User Address	String	100	NO
Mobile No	Mobile Number	String	12	NO

Table:3Login

Username	Username for Login	String	40	NO
Password	Password for Login	Integer	8	NO

5. COMPONENT DESIGN

- 1. Login:** In order for the user (patient & doctor) to login, a username and password needs to be entered and this was implemented in a form that contained fields for name and password, when the form is submitted, checks the database if the email is available and the validity of the password, if the password and username is valid the user is redirected to the homepage
-

else an error message is shown informing the user about the error. The login page also had a link that directs users that are not already registered to the registration page.

2. Registration: In the case of registration the user has to provide personal information like name, email, password , verify password ,date of birth, address, gender, telephone all these information are to be inputted into their respective fields in a form, when the form is submitted the first checks , if the password field is empty then if the password and verify password are the same,created by inserting all these information in the database else an error message is displayed.

3. Home page: This is the default page of the website i.e. the first page you see when you enter the app after you login or register, this page is the most important it implements some functionalities specified in the requirements like:

Search: A search box is provided at the top of the page, with this you can search for doctors by first name or last name, when the search is submitted the user is redirected to another page and a list of cars fitting that description is displayed. The homepage contains a slider displaying some of the services offered by the business, it also has a section briefly discussing information about the business, departments in the business and founders of the business.

4. Doctor: The doctor page displays the all the doctors available in the database, the doctor's information are displayed, also there is a book appointment button on each doctor row, when the button is clicked on the user is directed to another page where the patient can book that particular doctor appointments can only be booked if the user is logged in else the user is directed to the log in page after which the user is redirected back to the destined/original page.

5. Department: The department page displays all the necessary information about a particular department

available in the database, there is also a list of hospitals that have that particular department.

6. Appointment: The reservation page displays a form with the fields Hospital, Department, date when this form is filled and submitted by the users, the system uses this information to search through the database and displays doctors fitting the users specification, when the user selects a doctor, he is directed to a page to book the doctor, appointments can only be booked if the user is logged in else the user is directed to the log

in page after which the user is redirected back to the previous action page. After the user fills all necessary information and books the appointment a confirmation email is sent.

7. Contact: The contact page consists of a map displaying the company's location, also the address and contact details of the company and lastly a contact form in which users can fill and send in the

8. Account Profile: The account profile interface implements the following functionalities:

Edit profile: the user can view and edit their profile after which the new information is saved and updated in the database.

View appointments: all the bookings made by the user is displayed in a tabular form, appointment information is displayed.

Edit appointments: The user can change the dates of appointments; this new information is updated and stored in the database.

Cancel appointment: a field for booking reference is provided, with this booking reference the user's booking is canceled thereby deleting it from the database.

Logout: when this option is clicked the user is logged out and redirected to a more general homepage.

Admin module

1. Patient: In this interface the following functionalities are implemented for the administrator:

View patient: A list of all the patients in the database is displayed and when a patient item on the list is clicked, the patient details is displayed in a form

Add patient: the administrator can add patients to the database, a form is provided and the patient's details are entered, when the form is submitted the new patient is created and inserted into the database.

Edit patient information: The administrator can also edit patient information, whatever information entered is saved and updated in the database.

Delete patient: A delete option is placed beside each patients in the list of patients and when it is clicked that particular patient is deleted.

2. Doctor: In this interface the following functionalities are implemented for the administrator:

View doctor: A list of all the doctors in the database is displayed and when a doctor item on the list is clicked, the doctor details is displayed in a form

Add doctor: the administrator can add doctors to the database, a form is provided and the doctor's details are entered, when the form is submitted the new doctor is created and inserted into the database.

Edit doctor information: The administrator can also edit doctor information, whatever information entered is saved and updated in the database.

6. HUMAN INTERFACE DESIGN

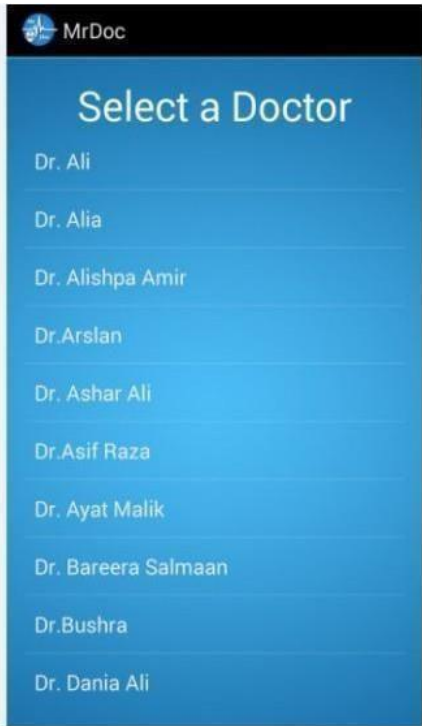
6.1. OVERVIEW OF USER INTERFACE

Describe the functionality of the system from the user's perspective. Explain how the user will be able to use your system to complete all the expected features and the feedback information that will be displayed for the user.

6.2. SCREEN IMAGES



Figure 2: The Signup screen



The list of doctors in the selected hospital

MrDoc

Logout

Add Doctor

View Doctors

View Patients

View Appointments

ADD DOCTOR

Name:

Address:

Mobile:

Email:

Specialty:

Date:

Submit

Our Contact

031 233 123

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Figure 25: The form to add a new doctor



16: The appointment reserved successfully

6.3. SCREEN OBJECTS AND ACTIONS

A discussion of screen objects and actions associated with those objects.

7. REQUIREMENTS MATRIX

REQ1. The application should enable patients and doctors to log in.

REQ2. The application should enable patients and doctors to register. REQ3.

The application should enable patients and doctors to log out.

REQ4. The application should allow Patients to book appointment. .

REQ5. The application should allow Patients to search for available doctors.

REQ6. The application should allow Patients to cancel their appointment.

REQ7. The system should allow Patients and Doctors to view and modify their profile.

REQ8. The system should allow Doctors to set their available time.

REQ9. The system should enable administrator to log in.

REQ10. The system should allow administrator to manage users .

REQ11. The system should allow administrator to delete past appointments.

REQ12. The system should allow administrator to manage & access Fire-base.

8. APPENDICES

Appendices may be included, either directly or by reference, to provide supporting details that could aid in the understanding of the Software Design Document.